TGS-2024043854 · DAY 2 · LAB 5 · 35 MIN

Multi-Agent Campaign Command Centre — Measure, Optimise & Report

Build a Human-AI Workforce with Autonomous AI Agents · Multi-agent (parallel swarm) · LO2

> Run three analysts in parallel, have a verifier check their numbers, and get a dashboard plus an optimisation plan you approve — then schedule it every Monday.

The business scenario

Week 1 of #ShiokColdBrew is over. Leadership wants to know: are we on track for 2,500 pre-orders, which posts work, what customers are saying and what competitors are doing? You launch a Kanban swarm — three parallel workers, a verifier that re-checks their numbers, and a synthesiser that builds the dashboard and recommendations. You decide which optimisations to apply.

You will deliver

- outputs/lab05/dashboard.html — KPI cards + charts (opens in a browser)
- outputs/lab05/optimisation-plan.md — 3 proposals with evidence, impact, risk
- outputs/lab05/verification.md — the verifier's re-calculation of the key numbers
- Your decision recorded on the swarm task
- A Monday 09:00 cron that delivers the weekly report to Telegram

Agents: 3 parallel workers + verifier + synthesiser (Kanban swarm)
You (the human): Spot-check one number yourself; approve which optimisations to run
Goal: Relate a parallel multi-agent design (workers + verifier + synthesiser) to speed, accuracy and cost, using metrics to verify the output and decide improvements (LO2).

Before you start

- Labs 3–4 complete: gateway running; researcher and auditor profiles exist
- A spreadsheet app to spot-check numbers

Lab files (data/)

File	What it is
`AGENTS.md`	Company context Hermes loads automatically
`brand-guide.md`	Brand voice, colours, platform formats

`campaign-brief.md`	Gula Melaka Cold Brew campaign brief (offer, KPIs, timeline)
`compliance-rules.md`	C1–C10 brand-safety and escalation rules
`product-catalogue.csv`	12 products and prices
`campaign-week1-metrics.csv`	Week-1 post metrics (12 posts)
`sales-by-outlet.csv`	Weekly sales for 12 outlets
`social-comments.csv`	20 social comments incl. complaints, legal, PII, spam
`competitors.csv`	6 fictional competitors
`tasks/swarm-definition-of-done.md`	Task brief (used with --body-file)
`assets/brand-palette.png`	Brand image asset
`assets/cold-brew-product-card.png`	Brand image asset
`assets/kopi-kaki-logo.png`	Brand image asset

All company data in this lab (Kopi Kaki Co., people, reviews, comments, metrics, competitors) is fictional and generated for training.

Steps

1. Add the analyst and strategist profiles

Reuse the researcher (competitor watch) and auditor (verifier) from Lab 4, and add two roles.

TERMINAL

```
cp -Rn data/* ~/kopi-kaki/
mkdir -p ~/kopi-kaki/outputs/lab05
hermes profile create analyst --clone --description "Computes marketing KPIs from CSV data with
exact formulas and shows its working"
hermes profile create strategist --clone --description "Synthesises analyses into an HTML
dashboard and an evidence-based optimisation plan"
hermes profile create listener --clone --description "Analyses customer comments for sentiment,
themes and risks; masks personal data"
```

2. Launch the swarm

A swarm creates parallel worker cards, then a verifier card, then a synthesiser card — all on the board, and workers start at once. So the definition of done goes INTO the goal and worker titles: every card points to ~/kopi-kaki/tasks/swarm-definition-of-done.md (formulas, output paths, charting rule). Note: the campaign data covers its first six days (17–22 Nov); the scenario is set in late Nov 2026, so web research must use the most recent real sources and label their dates.

TERMINAL

```
hermes kanban swarm "Week-1 #ShiokColdBrew review and optimisation plan. FIRST read ~/kopi-kaki/tasks/swarm-definition-of-done.md; data in ~/kopi-kaki; write outputs to ~/kopi-kaki/outputs/lab05/" \
  --worker "analyst:KPIs vs targets from ~/kopi-kaki/campaign-week1-metrics.csv and sales-by-outlet.csv per the definition of done" \
  --worker "listener:Sentiment and risk review of ~/kopi-kaki/social-comments.csv per the definition of done" \
  --worker "researcher:Competitor and cold-brew trend watch from the most recent real sources, dated and cited, per the definition of done" \
  --verifier auditor --synthesizer strategist
hermes kanban list            # copy the swarm root id
```

✓ Expected: 3 worker cards running in parallel, then verifier, then synthesiser.

3. Open the dashboard and plan

Read the KPI cards (pre-orders vs 2,500, engagement vs 8%, video views vs 40,000, cost per pre-order), the top and bottom posts and the three proposals.

TERMINAL

```
open ~/kopi-kaki/outputs/lab05/dashboard.html
```

✓ Expected: A self-contained HTML dashboard with charts and an optimisation plan.

4. Human verification — check one number yourself

Open campaign-week1-metrics.csv in a spreadsheet. Sum app_preorders and paid_spend_sgd, and compute cost per pre-order = spend ÷ pre-orders. Compare with the dashboard and with verification.md. If they differ, comment on the analyst task and ask for a correction — never forward a number you have not checked.

✓ Expected: Your figure matches (or you have sent a correction back).

5. Decide and record

PROMPT — PASTE INTO HERMES CHAT IN ~/KOPI-KAKI

```
I approve proposals <1 and 3> from outputs/lab05/optimisation-plan.md and reject proposal <2> because <reason>. Comment this decision on the swarm root card <swarm_root_id>, then update outputs/lab05/office-action-plan.csv for 24-30 Nov to apply the approved changes (status stays "proposed" so each action still requires my explicit review).
```

✓ Expected: Decision comment on the board and an updated office-action-plan.csv with action, owner, source evidence, due date and approval status.

6. Make it weekly

TERMINAL

```
hermes cron create "0 9 * * 1" "Weekly #ShiokColdBrew report: recompute KPIs vs targets from the
latest CSVs in ~/kopi-kaki, list the top and bottom 3 posts, summarise sentiment and escalations,
and recommend max 3 changes. Do not change any plan without my approval." --name weekly-campaign-
report --deliver telegram
hermes cron list              # copy the job id
hermes cron run <job_id>
```

✓ **Expected: A weekly report arrives on Telegram.**

Verification / checkpoint

- □ dashboard.html opens and shows the four KPIs vs target
- □ verification.md re-computes at least pre-orders, spend, engagement rate and cost per pre-order
- □ Your manual check matches (data covers 17–22 Nov)
- □ weekly-campaign-report is listed in `hermes cron list`

Checkpoint: Your own calculation of pre-orders and cost per pre-order matches the dashboard; the plan you approved is recorded on the task.

Troubleshooting

Symptom	Fix
Swarm workers can't find the CSVs	Say the absolute path (~/kopi-kaki/...) in the worker titles, or comment it on the root card.
Dashboard is blank	Ask the strategist for a single self-contained HTML file with inline SVG charts (no external scripts or data files).
Numbers disagree	That's why the verifier and you exist — request changes on the analyst card with the correct figure.

Stretch challenge

Feed the result back into Lab 3: ask Mia to add the approved changes to tomorrow's approval message, so the loop runs plan → publish → measure → improve without leaving Telegram.